

DESIGN YOUR CAREER IN VERY LARGE SCALE

VLSI technology will be propelled by wireless and multimedia communications for the next ten years, which will give the Indian VLSI/ASIC designing industry a fillip. So secure your job, position and moolah in this sector



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Are you a student of electronics engineering? If so, perhaps you have been exposed to the preliminary levels of integrated circuits (ICs) involving small-scale integration (SSI) circuits like logic gates or medium-scale integration (MSI) circuits like multiplexers and parity encoders. There is, however, a much bigger world out there involving miniaturisation at levels so great that even a micrometre or a microsecond is consid-

ered a huge measure! This is the world of VLSI—very large-scale integration. Let's figure out the opportunities in this field for professionals with subject-matter expertise in electronics.

Where is the demand?

Programmable logic devices, hardware description languages and design tools of today have changed the circuit design process to such an extent that it has become totally independent of the actual device manufacturing issues. According to India Semiconductor

Association (ISA), the Indian semiconductor and embedded design industry is expected to earn revenues of \$43 billion in 2015—a marked increase from the \$5.9 billion it earned in 2008.

In the present scenario, VLSI circuits are designed at one or more locations across the globe and finally manufactured at a different, distant location. This outsourcing aspect of VLSI design is attracting major international players in this field to India. Companies like Intel, Texas Instruments, IBM, Philips, Motorola, SGS-Thomson, Mentor Graphics and Cirrus Logic, to name a few, have already set up their design centres in India. India has some 9500 chip designers working for 300+ companies, and their ranks are swelling by 20 per cent a year. It has a unique advantage with respect to both the talent and market opportunities it provides.

The major recruiters in this fields are Texas Instruments, PMC Sierra, Infineon, Alliance Semiconductor, Analog Devices, Cadence, Synopsys, Mentor Graphics, Celox Networks, Cisco, Control Net, Cypress, DSPG, HCL, Intel, Lucent, Micron Tech, National Semiconductor, Motorola, Philips Semiconductor, Qualcomm, Sarsen, C2C, Atranta, Conexant, Moschip, Cradle Tech, Synplicity, STM, Paxonet, Wipro, TCS, eInfochips, Ishoni Networks and CGCoreEL.

What roles are in demand?

Though there is scope for fast growth in this area, the aspirants have to be competent, hardworking and constant